COKERNELS OF MAPS BETWEEN THE STANDARD PRESENTATIONS ARE AGAIN STANDARD PRESENTATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites five external items (a situation, several lemmas and an equation), all recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Finite subfamilies of a basis covering of a quasi-compact object. Stacks Project, Modules on Sites, Lemma lemma-silly-quasi-compact (tag 09B0). Produces the finite subset K' used to exhibit the presentation.
- The quasi-compact objects situation and its presentation equation. Stacks Project, Modules on Sites, Situation situation-quasi-compact-objects (tag 09AZ). Supplies the basis, the enough/ quasi-compact hypotheses and the standard presentation whose closure is proved.

Source reference. Original reference 0937: sources/stacks-chow/tags.txt, source lines 11505–11505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0938: sources/stacks-chow/tags.txt, source lines 11506–11506. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0939: sources/stacks-chow/tags.txt, source lines 11507–11507. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 093D: sources/stacks-chow/tags.txt, source lines 11511–11511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0G1W: sources/stacks-chow/tags.txt, source lines 20036–20036. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0937 assume (0938) and (0939) hold. Let $\mathcal{O}$ be a sheaf of rings. Then a cokernel of a map between modules as in (093D) is another module as in (093D).*

Proof. Let $\mathcal{F} = \mathrm{Coker}(\bigoplus j_{V_j!}\mathcal{O}_{V_j} \to \bigoplus j_{U_i!}\mathcal{O}_{U_i})$ as in (093D). It suffices to show that the cokernel of a map $\varphi : j_{W!}\mathcal{O}_W \to \mathcal{F}$ with $W \in \mathcal{B}$ is another module of the same type. The map φ corresponds to $s \in \mathcal{F}(W)$. Since $\bigoplus j_{U_i!}\mathcal{O}_{U_i} \to \mathcal{F}$ is surjective, by (0938) we may choose a covering $\{W_k \to W\}_{k\in K}$ with $W_k \in \mathcal{B}$ such that $s|_{W_k}$ is the image of some section s_k of $\bigoplus j_{U_i!}\mathcal{O}_{U_i}$). By (0939) the object W

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is quasi-compact. By Lemma 0G1W there is a finite subset $K' \subset K$ such that $\bigoplus_{k\in K'} j_{W_k!}\mathcal{O}_{W_k} \to j_{W!}\mathcal{O}_W$ is surjective. We conclude that $\mathrm{Coker}(\varphi)$ is equal to

$$\mathrm{Coker}\left(\bigoplus\nolimits_{k\in K'} j_{W_k!}\mathcal{O}_{W_k} \oplus \bigoplus j_{V_j!}\mathcal{O}_{V_j} \longrightarrow \bigoplus j_{U_i!}\mathcal{O}_{U_i}\right)$$

where the map $\bigoplus_{k\in K'} j_{W_k!}\mathcal{O}_{W_k} \to \bigoplus j_{U_i!}\mathcal{O}_{U_i}$ corresponds to $\sum_{k\in K'} s_k$. This finishes the proof. $\square$